



Hardik S. Mhatre

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CAREER FOCUS

Highly motivated and passionate chemical engineering student with a keen interest in research to solve real-world problems by applying my skills and knowledge.

EDUCATION

TECHNISCHE UNIVERSITÄT HAMBURG, *HAMBURG*, GERMANY

- **MS Chemical and Bioprocess Engineering** | ~2023
- Analysis and Design of Heterogeneous Catalytic Reactors
- Modern Methods in Heterogeneous Catalysis
- Applied Optimization in Energy and Process Engineering
- Applied Thermodynamics for industrial applications

DATTA MEGHE COLLEGE OF ENGINEERING, *NAVI MUMBAI*, INDIA

- **BE Chemical engineering** | ~2019 – 2023
 - Study of recent development and drying kinetics of different dryers (Spray dryer).
 - CGPA: **8.52**
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PROFESSIONAL EXPERIENCES

Tesa SE

September 2024-Present

Working student (R&D)

Development and testing of pressure-sensitive adhesives at Tesa SE, including scaling up PSA film production from lab scale to commercial scale and conducting various tests to ensure quality and performance during the scale-up process. – Acrylic & Rubber adhesive, Polymers, DOE, Formulation, LIMS.

Institute für Thermische Verfahrenstechnik

March 2024-August 2024

Student Research Assistant

Engineered a predictive model in Python for ultra-high pressure liquid densities by bridging the Tait and Phonon Theory with fundamental molecular and thermodynamic properties. Involving rigorous data comparison against TREND EOS and parameter optimization. – Data analysis, Python, Mathematical Model

Indian Institute of Technology Bombay

June 2023 – July 2023

Research Internship

Worked to determine the synergistic effect of graphene, nickel, and Mxene as a catalyst on the reaction rate kinetics such as rate constant and reaction order of Mg-based metal hydride nanocomposites for solid-state hydrogen storage using nonlinear regression – TPR-TPD, XRD, SEM, Raman spectroscopy, FTIR, Kinetics.

Institute of Chemical Technology

August 2022 – April 2023

Dissertation project

The dissertation project based on optimization of spray drying parameters such as residence time for drying milk and design of various mechanical parts of spray dryer such as atomizer, air distribution plate and cyclones. This also led us to hands on operation of lab scale spray dryer. – Spray dryer operation, Spray dryer Design, Optimization.

Hindustan Petroleum Corporation Limited

July 2022 – August 2022

Chemical engineering Internship (Operations)

Worked in the new fluidized bed catalytic cracking unit's reactor regenerator and also got a deeper insight into troubleshooting the FGTU (Fuel Gas Treatment Unit) amine absorber – Plant operations, DCS, Line tracing, Pump troubleshooting, Shell and tube heat exchanger troubleshooting, FCC Reactor, Process optimization.

Institute of Chemical Technology
Research Internship

June 2022 – July 2022

Worked on a research project funded by Hindustan Unilever to study the degradation kinetics and stabilization of hydrogen peroxide in a surfactant solution – HPLC, GC, UV-Vis spectroscopy, Krüss K-11 Tensiometer

LICENSES & CERTIFICATIONS

- Aspen plus simulation software.
- Process Safety (Dow)

PROJECTS

- Autonomous Multi Objective Optimization of SNAr Reaction system using a capillary flow reactor
- Real time hydrodynamic testing and mathematical model Development of a Coiled flow inverter (CFI) bioreactor for cell population heterogeneity (GitHub: github.com/Hardik-Ma/Real-time-Reactor-Hydrodynamics-analyzer)
- Simulation of gas separation membrane for hydrogen recovery from purge gas in e-methanol production (GitHub: github.com/Hardik-Ma/H2_Gas_Membrane_Simulation)
- Adsorption & Membrane Cascade Modelling for Biogas Purification | Python (GitHub: <https://github.com/Hardik-Ma/Biogas-Purification>)
- Recovery of precious and heavy metals by hydrometallurgical leaching of printed circuit boards.
- Production of biofuel and ethylene from lignocellulosic biomass.
- Process simulation of biomethanol from lignocellulosic biomass gasification using a downdraft gasifier.

PUBLICATIONS

- Published conference paper “Fabrication and characterization of electric double layer supercapacitor based on activated carbon electrodes in aqueous electrolytes.” in ICATE-2020 (4th International Conference on Advanced trends in Engineering) ISBN:978-93-5391-503-2.

ACHIEVEMENTS

- Third position in European Committee for the Use of Computers in Chemical Engineering Education (EURECHA) Process Design Contest 2025

TECHNICAL SKILLS

Aspen Plus, COMSOL Multiphysics, DWSIM, Python, C, MS. Excel, MS. Office, Autocad 3D, Matlab, GAMS

LANGUAGES

English C1, German B2.